Biology Class 11 National Book Foundation FBISE Quality Notes
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Chapter 12: Inheritance

Heredity, variation and the gene (SLO 1)

- ***Heredity:*** *similar characteristics passed from parents to offspring.* ***Variations:*** *differences among offspring and parents.* ***Genetics:*** *science of heredity and variation (study of genes).*

Gene: *A basic unit of heredity: a small segment of DNA made of a sequence of nucleotides, which codes for the formation of a polypeptide.*

- *Each gene has two polynucleotide chains; one chain, the* ***template****, determines the protein. Three bases code for one amino acid.* ***Transcription*** *(nucleus): the template is copied into mRNA (codons).* ***Translation*** *(cytoplasm): the ribosome translates mRNA into a protein.*

12.1 Mendelian inheritance (SLO 2)

- *Genetics began in 1900 with rediscovery of an article published in 1866 by* ***Gregor Johann Mendel****, an Augustinian monk, the father of genetics. He worked on pea plants.*
- ***Steps:*** *(1) develop* ***true (pure) breeding*** *varieties by repeated self-fertilization (offspring always of its own phenotype); (2)* ***hybridization****: cross-fertilization between plants with contrasting phenotypes. He studied* ***seven pairs of contrasting traits*** *and followed one trait, then two traits (the secret of his success).*

Term	*Meaning*
Allele	*Alternative form of a gene on the same locus (factor).*
Genotype / phenotype	*The pair of alleles / the expressed trait.*
Dominant / recessive	*Factor expressed in F1 / factor that fails to express (capital / small letter).*

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Term	Meaning
Homozygous / heterozygous	Both alleles same (RR, rr; homozygote) / different (Rr; heterozygote).
P, F1, F2	Parental generation; first filial generation; second filial generation.
Monohybrid / dihybrid cross	Cross between individuals differing in one trait / two traits.

Monohybrid cross - Law of segregation

- Round (RR) x wrinkled (rr): F1 all round (Rr); F1 self-fertilized: F2 **3 round : 1 wrinkled** (also genotypic ratio **1 RR : 2 Rr : 1 rr**) for all seven traits. In F3, 1/3 of F2 round bred true, 2/3 gave 3:1 again, and F2 wrinkled bred true.

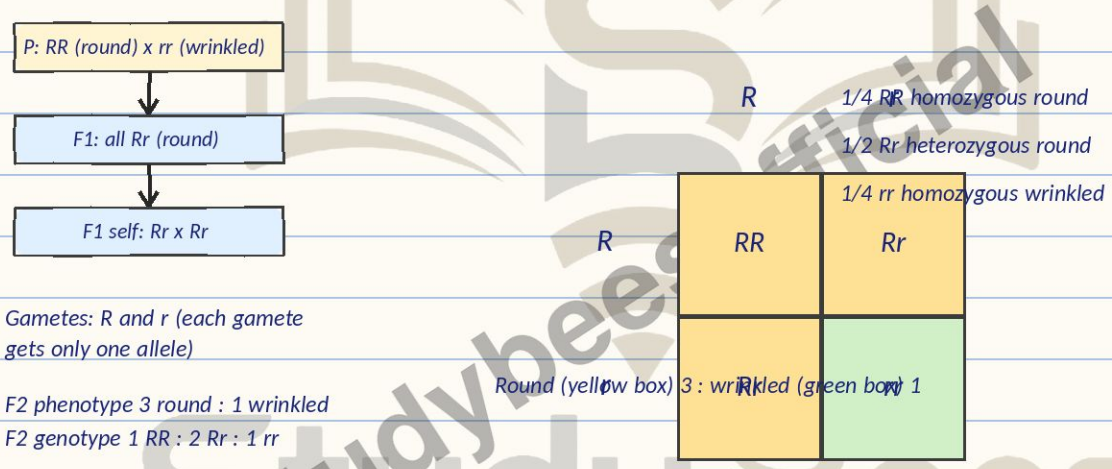


Fig: Monohybrid cross (Law of segregation)

Law of segregation: The two coexisting alleles for each trait in an individual segregate (separate) from each other at meiosis so that each gamete receives only one of the two alleles; alleles unite again at random at fertilization.

Dihybrid cross - Law of independent assortment

- Round yellow (RRYY) x wrinkled green (rryy): F1 all round yellow (RrYy). F1 self: F2 has four phenotypes **9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green**; recombinant types (round green, wrinkled yellow) were surprising. F1 makes four gamete types RY, Ry, rY, ry in equal amounts: 4 x 4 =

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16 combinations.

	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

P: RRYY (round yellow) x rryy (wrinkled green)

F1: all RrYy (round yellow, dihybrid)

F1 x F1: 4 gamete types RY, Ry, rY, ry (equal)

16 equally likely combinations in F2:

- 9 round yellow
- 3 round green
- 3 wrinkled yellow
- 1 wrinkled green

Phenotype ratio 9 : 3 : 3 : 1

Each pair of alleles assorts independently

Fig: Dihybrid cross (Law of independent assortment)

Law of independent assortment: *Each pair of alleles assorts independently of other pairs of alleles during gamete formation; alleles of each pair of contrasting traits have equal probability of assorting with alleles of the other pair.*

- *R. C. Punnett (Cambridge) devised the **Punnett square** to summarize fusion of gametes in genetic crosses.*

Independent assortment and chromosomes (SLO 3, 4, 5)

- *Genes concerned must be on **different chromosomes** (seed colour and shape are on different pairs). At metaphase I each homologous pair lines up independently of the others (**random orientation**), so maternal and paternal homologues are sorted into daughter cells independently. Humans: 2^{23} = about **8 million** possible chromosome combinations in gametes.*
- ***Limitations:** applies to diploid organisms only; genes on the **same chromosome** (linked) do not assort independently unless crossing over occurs;*

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*genes on the **X chromosome** give a different pattern (males show recessive X-linked traits more often as they have one X).*

- ***Usefulness:** with mutation and crossing over, a major source of variation; new combinations are needed for adaptation in changing environments. Meiosis gives genetically distinct gametes and random fertilization gives different zygotes. **Three sources of variation:** crossing over (prophase I), random assortment (metaphase I), random fusion of gametes.*

Inheritance and probability (SLO 6)

***Probability:** The chance that an event occurs: P = a / n (favourable cases / total possible cases). Coin toss: P = 1/2.*

- *Segregation and re-pairing of alleles obey probability rules. Monohybrid cross: probability of dominant phenotype in F1 is 100% and recessive 0%; in F2 dominant 3/4 and recessive 1/4.*
- ***Product rule:** probability of two independent events together = product of individual probabilities. Round yellow in F2 = 3/4 x 3/4 = **9/16**.*

12.2 Exceptions to Mendelian inheritance (SLO 7, 8, 9)

- ***Dominance relations:** relationship between contrasting alleles at the same locus in the heterozygote. Mendel saw only **complete dominance**; later exceptions (non-Mendelian): incomplete dominance, co-dominance, multiple alleles.*

Incomplete dominance - four o'clock plant (SLO 8)

- *Carl Correns (1899), **Mirabilis jalapa**: red x white gives F1 all **pink**; pink x pink gives **1 red : 2 pink : 1 white** (phenotype ratio = genotype ratio). Heterozygote is intermediate: neither allele fully dominant. Only capital letters with dashes are used (RR, R'R', RR').*

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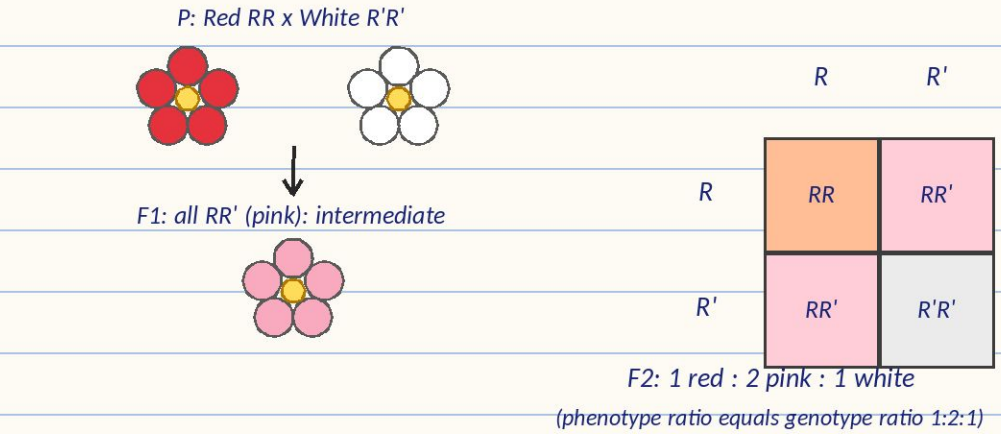


Fig: Incomplete dominance in four o'clock (Mirabilis jalapa)

Co-dominance (SLO 9)

- *Both alleles at the same locus express independently in the heterozygote.* ***MN blood group:*** *alleles L^M and L^N: L^ML^M = M antigen only; L^NL^N = N only; L^ML^N =* ***both*** *M and N.*

Incomplete dominance	**Co-dominance**
Heterozygous: both genes blend their effects	Heterozygous: both genes express independently
Heterozygote shows intermediate phenotype	Heterozygote shows both parental phenotypes at a time
Flower colour of four o'clock	Human MN and AB blood groups

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Alleles and multiple alleles (SLO 10)

- ***Alleles:*** *alternative forms of a gene at the same locus on homologous chromosomes.* ***Multiple alleles:*** *more than two allelic forms of a gene, produced by* ***mutation****, all at the same locus; a diploid individual carries only two of them (e.g. A_1, A_2, A_3: six genotype combinations). Some traits have 100 or more. ABO blood groups are an example.*

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12.3 Blood group systems (SLO 11-16)

- Up to **33 blood group systems** with over 300 antigens (International Society of Blood Transfusion); antigens are mostly **glycoproteins** on the RBC surface. Important: ABO, MNS, Rhesus, Lutheran, Kell, Lewis, Duffy, Kidd. **ABO and Rh** are the most significant; over 200 minor groups (rare types) seldom complicate transfusions.

ABO system (SLO 11, 12)

- Discovered by **Karl Landsteiner** (1901); also in apes, chimpanzees, bonobos, gorillas. Based on antigens A and B (glycoproteins). Antigen A = type A; B = type B; both = AB; none = O.
- **Genetics:** single autosomal gene **I** (isohaemagglutinogen) with three alleles I^A (makes A), I^B (makes B) and **i** (neither). I^A and I^B are completely dominant over i; I^A and I^B are **co-dominant** (I^AI^B = AB).

Blood group	Genotype	Antigen on RBC	Antibody in plasma
A	I^AI^A or I^Ai	A	Anti-B
B	I^BI^B or I^Bi	B	Anti-A
AB	I^AI^B	A and B	None
O	ii	None	Anti-A and anti-B

- Alleles express from the early embryo until death, so blood group never changes. ABO antibodies need no stimulus; they appear from embryonic life. Any person above 6 months has clinically significant anti-A / anti-B. Antigens are also secreted in saliva, tears and urine by **secretors** (dominant gene Se, chromosome 19).
- **Bombay phenotype:** expression of ABO also depends on gene **H** (chromosome 19) making H substance, the precursor of A and B antigens. Persons **hh** make no H substance so no A or B antigen: appear as O. First described in Bombay (India).

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Transfusion principle and universal donor / recipient (SLO 14)

- *In wrong transfusion, antigens of the **donor** react with antibodies (agglutinins) of the **recipient**: RBCs clump = **agglutination**. So transfusions follow donor antigens and recipient antibodies. A receives A or O; B receives B or O; AB receives any (universal recipient: no antibodies); O gives to all (universal donor: no antigen).*

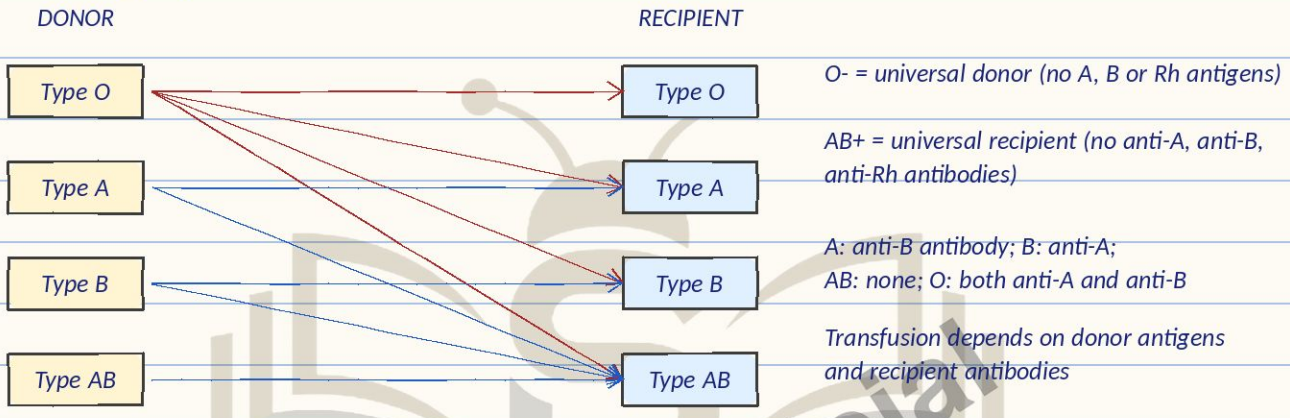


Fig: ABO transfusion compatibility

- ***O negative** is the actual universal donor (no ABO or Rh antigen); **AB positive** the actual universal recipient (neither anti-A, anti-B nor anti-Rh).*
- ***Alternatives to transfusion:** volume expanders (normal saline, Ringer's solution), haematopoietic growth factors, erythropoietin (kidney hormone, makes more RBCs), aprotinin (reduces bleeding in heart surgery).*
- ***Why is allele i frequent?** ii x ii gives all O; heterozygous crosses (I^Ai x I^AI^B, etc.) also give O offspring, so i persists (O frequency: Australia 40%, Canada 39%, Iceland and Ireland 47%, UK and USA 37%).*
- ***Blood donation:** a social service; one unit of blood can save three lives; Quran Surah 5 verse 32: saving a life is as saving all mankind; voluntary unpaid donors keep the blood supply safe.*

Other system	Notes
MNS	Autosomal, chromosome 4; co-dominant alleles L^M, L^N; anti-M / anti-N rarely cause transfusion reactions.
Lutheran	Four pairs of allelic antigens (chromosome 19); antibodies rare, not clinically significant.

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Other system	Notes
Kell	Anti-K first noticed in Mrs Kellacher; 25 Kell antigens; causes haemolytic disease of newborn and transfusion reactions.
Duffy	Fy glycoprotein (Fya, Fyb): four phenotypes; IgG antibodies; transfusion reactions.
Kidd	Jk glycoprotein, urea transporter; antibodies rare but severe reactions.

Rh blood group system (SLO 16, 17)

- Named from **Rhesus monkey** (Landsteiner, 1930s). The **D antigen** = Rh factor. Present = **Rh positive**; absent = **Rh negative**. Gene D (completely dominant) forms D antigen; allele d does not. DD or Dd = Rh+; dd = Rh-.
- **Anti-Rh antibody** is made only in Rh-negative blood and **needs a stimulus** (exposure to Rh antigen), unlike ABO antibodies; once stimulated it continues for life. Rh+ blood to an Rh- recipient is incompatible; Rh- blood (never exposed) can go to Rh+.

Erythroblastosis foetalis (SLO 17, 18, 19)

- Develops when the mother's anti-Rh antibodies cross the placenta into an Rh-positive foetus: **haemolysis** of foetal RBCs; anaemia; the foetus releases many immature **erythroblasts** (hence the name); abortion or stillbirth possible; liver and spleen swell; **bilirubin** damages brain and causes **jaundice**. A baby born alive needs immediate replacement by Rh- blood free of anti-Rh antibodies.
- **Cause:** Rh-negative woman (dd) marries an Rh-positive man and conceives an Rh+ baby (maternal-foetal Rh incompatibility). If the father is DD all children are Dd (Rh+); if Dd half are Dd. The first baby is usually safe; later Rh+ babies are at risk.

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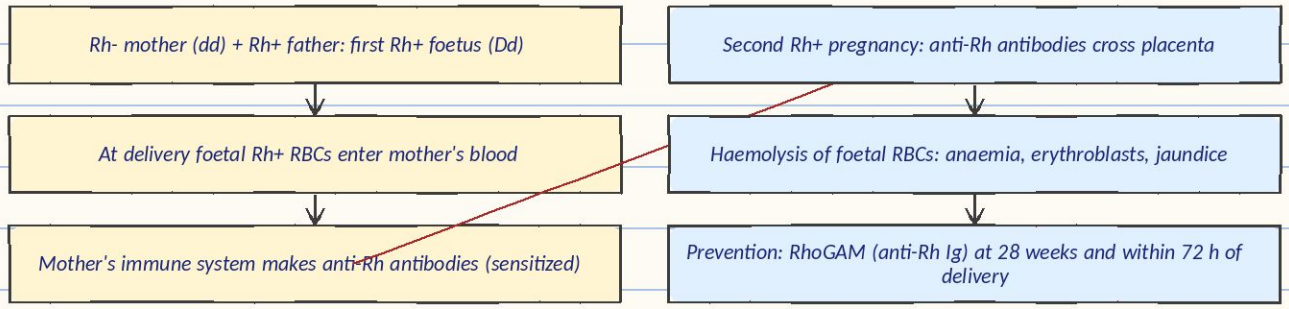


Fig: Rh incompatibility and erythroblastosis foetalis

- ***Prevention:*** *Rh immunoglobulin (**RhIg, RhoGAM**) containing anti-Rh antibodies destroys foetal RBCs in the mother's blood before she makes her own antibodies. Given at **28 weeks**, within **72 hours of delivery** of an Rh+ baby, within 72 hours of miscarriage, abortion or ectopic pregnancy, after amniocentesis / CVS and after vaginal bleeding. It does not help women already sensitized.*

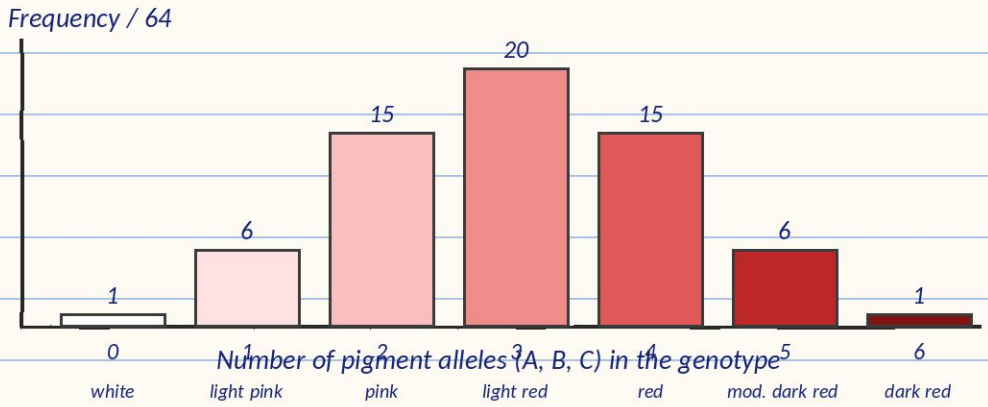
12.4 Polygenic inheritance and epistasis (SLO 20-23)

- ***Polygenic (quantitative) traits*** *show **continuous variation** (height, weight, intelligence, skin colour, wheat grain colour). They are encoded by alleles of two or more gene pairs at different loci acting **additively**; the genes are **polygenes** with small positive or negative effects.*

Wheat grain colour (SLO 21)

- *Dark red (AABBCC) x white (aabbcc): F1 light red (AaBbCc, looks like incomplete dominance); F2 has **seven shades** in ratio **1 : 6 : 15 : 20 : 15 : 6 : 1**. Three gene pairs (Aa, Bb, Cc) contribute; A, B, C each add an equal dose of red pigment; a, b, c add none. 6 alleles = dark red, 0 = white; 1 = light pink, 2 = pink, 3 = light red, 4 = red, 5 = moderately dark red. Light, water and nutrients also affect colour.*

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AaBbCc x AaBbCc: F2 ratio 1 : 6 : 15 : 20 : 15 : 6 : 1 (7 shades, bell curve)

Fig: Polygenic inheritance of wheat grain colour (also human skin colour)

Human skin colour (SLO 21)

- Determined by **melanin**; at least three genes A, B, C with no complete dominance. Dark-skin alleles A, B, C add melanin; light-skin alleles a, b, c do not. Seven shades from very light (aabbcc) to very dark (AABBCC); most people AaBbCc (e.g. a mulatto, offspring of black and white parents).
- **Polygenic traits in humans (SLO 22):** height, skin colour, risk of disease and resistance, intelligence, blood pressure, autism, longevity, cancer, heart disease, diabetes, bipolar disorder.

Epistasis (SLO 23)

Epistasis: Non-allelic interaction in which the effect of a gene at one locus interferes with or hides the effect of a gene at another locus. The suppressing gene = epistatic (inhibiting) gene; the suppressed gene = hypostatic gene.

Dominance	Epistasis
Involves a single pair of alleles	Involves two pairs of alleles (different loci)
A gene suppresses expression of its own allele	A gene suppresses expression of an allele of another gene
Only the recessive allele is suppressed	Suppresses both dominant and recessive alleles of another gene
Effect only due to dominant allele	Both dominant and recessive allele can be epistatic

- **Epistasis vs polygenic:** both involve more than one gene; epistasis gives

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discontinuous *variation, polygenic gives* ***continuous*** *variation.*

- ***Labrador coat colour (recessive epistasis, 9:3:4):*** *B (black) dominant to b (chocolate); a second locus E / e:* ***ee*** *makes the coat* ***yellow*** *regardless of B or b (ee epistatic). BBEE x bbee: F1 all BbEe (black); BbEe x BbEe:* ***9 black : 3 chocolate : 4 yellow.***

	BE	Be	bE	be
BE	BBEE	BBEe	BbEE	BbEe
Be	BBEe	BBee	BbEe	Bbee
bE	BbEE	BbEe	bbEE	bbEe
be	BbEe	Bbee	bbEe	bbee

BBEE (black) x bbee (yellow): F1 all BbEe (black)

BbEe x BbEe gives three coat colours:

- 9 black (B_E_)
- 3 chocolate (bbE_)
- 4 yellow (__ee)

Ratio 9 : 3 : 4 (recessive epistasis)

ee is epistatic: masks B and b (different loci)

Foxglove (dominant epistasis): 12 : 3 : 1

Fig: Epistasis - coat colour in Labrador retriever

- ***Foxglove (dominant epistasis 12:3:1):*** *d (light red) / D (dark red) pigment; W restricts pigment to the throat (epistatic): dominant W masks colour:* ***12 white (pigment in throat only) : 3 red : 1 light red.***
- ***Sweet pea (Lathyrus odoratus):*** *two genes both needed for purple pigment; a precursor is converted by two enzymes. If either gene is homozygous recessive no colour forms: ratio 9 purple : 7 white (standard).*
- *The Bombay phenotype (hh hides ABO) is also an example of epistasis.*

12.5 Gene linkage and crossing over (SLO 24, 25, 26)

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Gene linkage: *The phenomenon of genes on the same chromosome staying together and being inherited together. Linked genes on an autosome = autosomal linkage; on a sex chromosome = sex linkage.*

- *Each chromosome carries hundreds to thousands of genes. All linked genes on one homologous pair form a* ***linkage group****, so the number of linkage groups = the number of homologous pairs. Linked genes do not assort independently so the ideal Mendelian ratio is deviated.*
- ***Detection: dihybrid test cross*** *(heterozygous F1 x recessive parent): equal* ***1:1:1:1*** *= no linkage; more parental and fewer recombinant types =* ***incomplete linkage****; only parental types =* ***complete linkage*** *(dihybrid F2 gives 3:1 instead of 9:3:3:1).*
- ***Morgan's experiment*** *(****Drosophila melanogaster****, about 85 contrasting trait pairs): long wing (Vg) dominant to vestigial (vg); broad abdomen (A) dominant to narrow (a). Long-broad x vestigial-narrow: F1 all long-broad; F2 about 3/4 long wing broad abdomen and about 1/4 vestigial narrow: genes on the same chromosome, no recombinants: 9:3:3:1 became 3:1.*
- ***Crossing over:*** *exchange of parts between maternal and paternal chromatids while homologues are paired in prophase I; recombinant chromatids give new allele combinations in gametes: genetic variability; it* ***modifies the progeny****.*

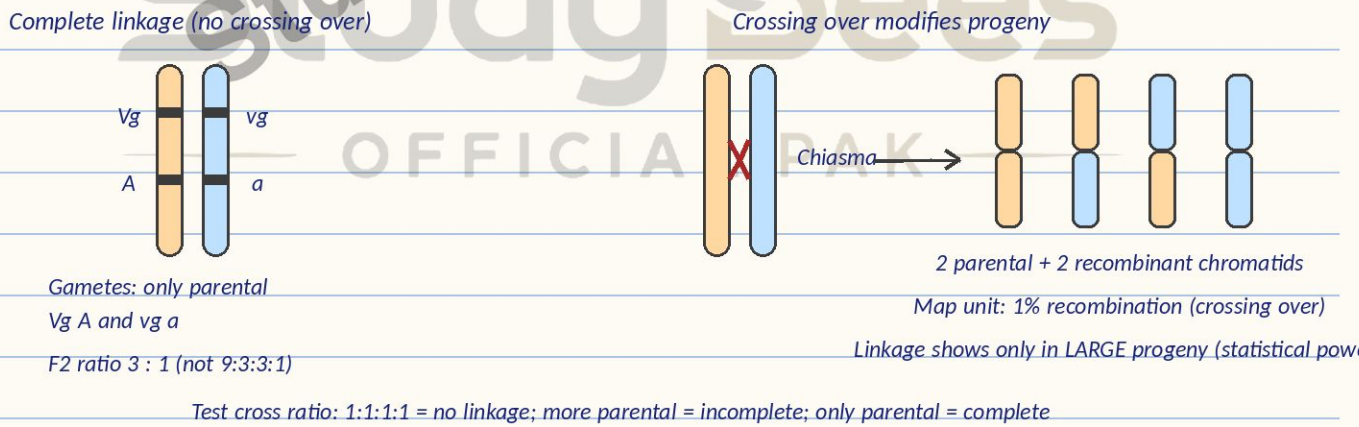


Fig: Gene linkage and crossing over (Morgan, Drosophila)

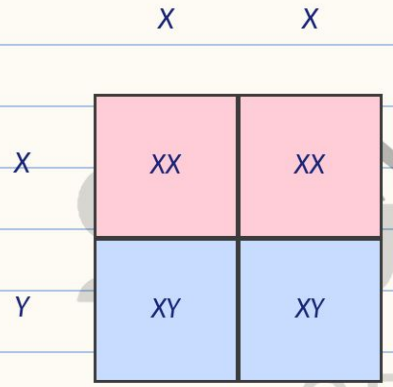
- ***Large progeny needed (SLO 26):*** *Morgan: linkage depends on the* ***distance*** *between genes; crossing over frequency is governed by distance. A* ***map unit*** *=*

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1% recombination. A large sample gives more statistical power to see consistent patterns (deviation from independent assortment); a small group shows mixing of traits.

12.6 Sex determination (SLO 27, 28, 29)

- *Sex is determined by **sex chromosomes**; the rest are **autosomes**. **Karyotype** = arrangement of chromosomes. **Drosophila:** 4 pairs (3 autosomal + sex pair); female XX (identical, rod-shaped), male XY (X rod-shaped, Y hooked). **Humans:** 23 pairs: 22 autosomal + 1 sex pair; female XX, male XY. The **SRY gene** (sex-determining region Y), only on Y, makes a person male; a person with XY but without it would be female.*
- ***XX-XY pattern:** male is **heterogametic** (X or Y sperm in equal numbers); female is **homogametic** (X eggs only). Sperm decides sex: X sperm gives girl, Y sperm gives boy; sex ratio 1:1.*



Female XX: homogametic (one type of egg, X)
Male XY: heterogametic (X and Y sperms, 1:1)
Sperm decides sex: X sperm = girl, Y sperm = boy
Sex ratio 1 : 1 (50% chance each pregnancy)

Humans: 22 pairs autosomes + 1 pair sex chromosomes
Y carries SRY (sex-determining region Y) gene
Drosophila: 4 pairs (3 autosomes + XX / XY)

Fig: XX-XY sex determination

12.7 Sex linkage (SLO 30-34)

- *Traits controlled by genes on sex chromosomes = **sex-linked**. **X-linked** (only on X), **Y-linked** (only on Y), **X-Y linked / pseudoautosomal** (alleles on both X and Y; autosomal pattern). Wild type = normal; mutant = not normal.*
- ***Morgan's Drosophila eye colour:** white-eyed male x red-eyed female: F1 (1237) all red; F2: 3470 red : 782 white, and **all white-eyed were males**. A test*

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cross of F2 red female x white male gave 129 red females, 132 red males, 88 white females, 86 white males (proves white eye does not kill females). White female x red male: all daughters red, all sons white. Conclusion: ***white-eye gene is on the X chromosome and recessive; Y has no allele****. Genotypes:* X^RX^R, X^RX^r, X^RY *(red);* X^rX^r, X^rY *(white).*

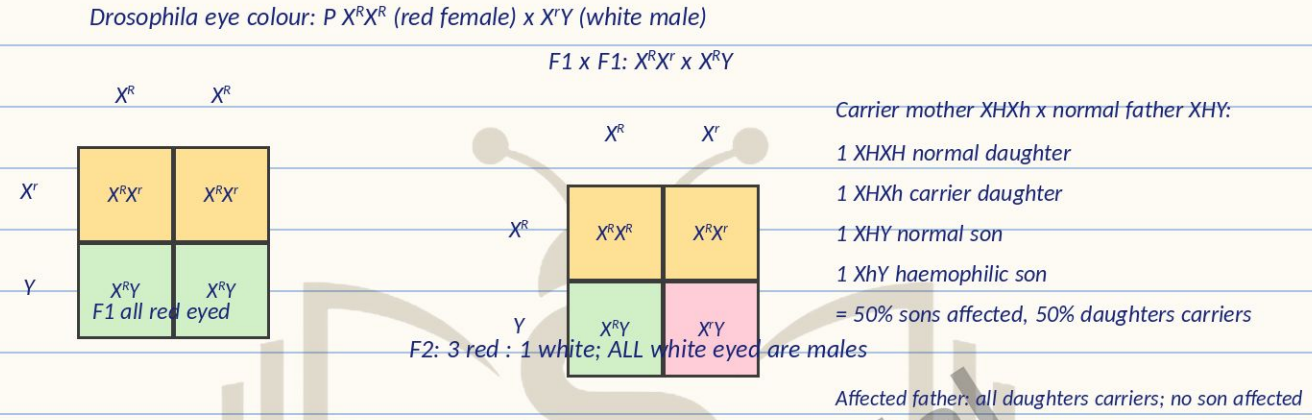


Fig: Sex-linked (X-linked recessive) inheritance

Pattern	Features
X-linked recessive	Carrier female (one mutation) usually unaffected; **all males with the mutation are affected**; carrier mother: 50% of children inherit it; **all daughters of an affected father are carriers; no son of an affected father is affected**; traits "zigzag" from maternal grandfather through carrier daughter to grandson. Examples: haemophilia A, B; colour blindness; testicular feminization.
X-linked dominant	Affected mother: 50% of sons and daughters affected; affected father: all daughters affected, no sons. Example: hypophosphatemic (vitamin D resistant) rickets.
Y-linked (holandric)	Genes only on non-homologous part of Y; only males show them; father to son. Examples: hypertrichosis (hair on pinna rim), porcupine man, webbed toes; also TDF (testis determining factor) and H-Y gene.

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- *Experimental mating is impossible in humans, so inheritance is traced by **pedigree analysis** using agreed symbols.*

Haemophilia (SLO 34)

- *Rare X-linked recessive: blood fails to clot (reduction, malfunction or absence of clotting factors); may bleed to death from minor cuts. Types: **A** (factor VIII; 80%), **B** (factor IX; 20%) are X-linked recessive; **C** (factor XI) is autosomal recessive (both sexes equally). A and B affect men twice as often; a woman needs two recessive alleles (X^hX^h), a man only one. Genes: H normal, h haemophilia. It never passes directly father to son. A father and son can both be affected only if the mother is a carrier (or affected).*
- ***Treatment:** mild type A: **desmopressin** (synthetic hormone stimulating factor VIII); type A: synthetic factor VIII (octocog alfa); type B: engineered factor IX (nonacog alfa). Normal clotting: platelets clump; clotting factors in a cascade convert fibrinogen (factor I) to fibrin which traps blood cells (13 factors, I to XIII).*

Colour blindness

- *Normal trichromatic vision: three cone types (red, green, blue) with **opsins**. Genes for red and green opsins are on the X chromosome; blue opsin on autosome 7. **Dichromacy:** one primary colour unseen (protanopia = red blind; deuteranopia = green; tritanopia = blue). **Monochromacy** (true colour blindness): only one colour seen; blue cone monochromacy is X-linked recessive (red and green cones absent) = **red-green colour blindness**. More common in men (twice as likely); zigzag inheritance, never directly father to son.*
- ***Sex limited traits:** expressed in only one sex (e.g. milk yield, beard growth). **Sex influenced traits:** dominant in one sex and recessive in the other (e.g. baldness: dominant in men, recessive in women).*

Embryonic screening (SLO 35)

- *Checks an embryo for genetic disease alleles and chromosomal / gene mutations.*

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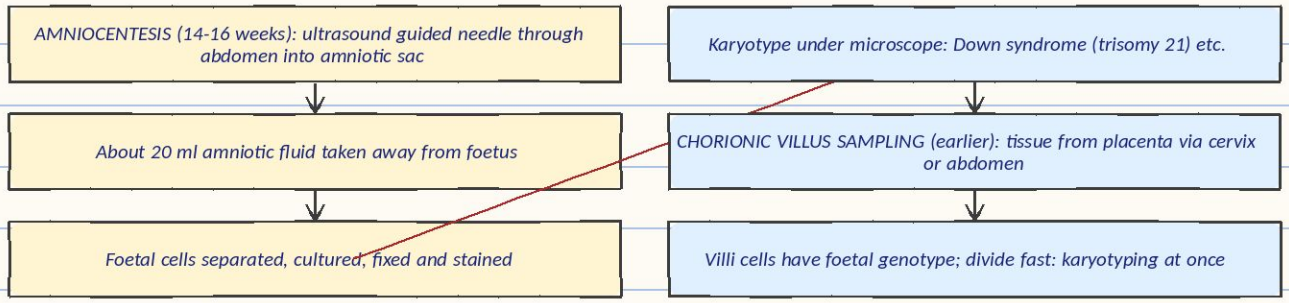


Fig: Embryonic screening techniques

- ***Amniocentesis:*** *local anaesthetic; ultrasound guided needle through abdominal wall and uterus into the amniotic sac, away from the foetus; about* ***20 ml amniotic fluid*** *removed; foetal cells separated, cultured, fixed, stained; chromosomes examined, e.g.* ***Down syndrome (trisomy 21).***
- ***Chorionic villus sampling (CVS):*** *tissue from the placenta (chorionic villi: fingerlike, same genetic material as foetus) taken through the cervix (transcervical) or abdomen (transabdominal); cells divide fast so karyotyping can be done immediately; safe, earlier diagnostic prenatal test.*

Genetic problems with solutions

Problem	Answer
12.1 PKU and albinism (unlinked recessives); both parents heterozygous for both	PKU 1/4; albinism 1/4; both 1/4 x 1/4 = 1/16
12.2 CC x Cc	1/2 CC, 1/2 Cc, no cc
12.3 Tall yellow x tall yellow gives dwarf green	Parents TtYy x TtYy; tall yellow offspring: TTYY, TTYy, TtYY or TtYy
12.4 TtYy self: (a) tall green? (b) tall yellow is TTYY?	(a) 3/4 x 1/4 = 3/16; (b) 1/9
12.5 Pink x red four o'clock	1/2 red, 1/2 pink; pink : red = 1 : 1
12.6 Parents A and B; four children with four blood groups	I^Ai x I^Bi gives AB, A, B and O

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Problem	Answer
12.7 Mother B, child O	Mother I^Bi; child ii; father cannot be I^AI^A, I^BI^B or I^AI^B
12.8 Rh- woman, Rh+ man whose father was Rh-	Man Dd, woman dd; half the children Dd (Rh+) at risk of erythroblastosis (mainly after first pregnancy)
12.9 Two chocolate Labradors give a yellow puppy	Both bbEe; yellow (ee) again 1/4
12.10 Boy or girl 50:50	Man makes equal X and Y sperms; the sperm decides sex
12.11 Four daughters; fifth child a son?	1/2 (50%): each pregnancy is independent
12.12 Both parents A, child O: A daughter or A son next?	I^Ai x I^Ai: A = 3/4; A daughter = 3/4 x 1/2 = 3/8; A son 3/8
12.13 Normal parents haemophiliac child?	Yes: X^HX^h x X^HY; 1/4 of children (1/2 of sons) haemophiliac
12.14 Carrier woman x normal man: colour-blind son	1/4 of all children (1/2 of sons)
Test cross of round pea of unknown genotype	Cross with wrinkled rr: all round = RR; 1 round : 1 wrinkled = Rr
Four o'clock: red x red; red x pink; pink x white; white x white	All red; 1 red : 1 pink; 1 pink : 1 white; all white
Man type A x woman AB	Children A, AB or B (if man I^Ai); A or AB (if I^AI^A); his parents / children tell his genotype
Two white sheep give a black offspring	Both heterozygous (white dominant); next lamb black 1/4
Normal parents, albino child: next child normal?	Both Aa; normal 3/4

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Chapter 12: Short Questions with Answers (SLO based)

Q1. [SLO 1] Define heredity, variation and genetics.

Ans: Heredity: passing of similar characters from parents to offspring. Variation: differences among offspring and parents. Genetics: science of heredity and variation.

Q2. [SLO 1] Define gene.

Ans: A basic unit of heredity, a small segment of DNA with a nucleotide sequence that codes for a polypeptide.

Q3. [SLO 1] What are transcription and translation?

Ans: Transcription: copying of the template chain of a gene into mRNA in the nucleus. Translation: ribosomes make protein from mRNA codons in the cytoplasm.

Q4. [SLO 2] Why is Mendel called the father of genetics?

Ans: He first explained the mechanism of inheritance by careful experiments on pea plants; his work is the foundation of genetics.

Q5. [SLO 2] What are true breeding plants?

Ans: Plants that produce offspring of their own phenotype on self-fertilization.

Q6. [SLO 2] Why did only about one fourth of Mendel's F2 plants show the recessive trait?

Ans: F1 (Rr) gamete alleles segregate; only fusion of two r gametes (1/2 x 1/2) gives rr, so 1/4 are recessive.

Q7. [SLO 2] State the law of segregation.

Ans: The two alleles of a trait separate at meiosis so each gamete receives only one; they unite at random in fertilization.

Q8. [SLO 2] State the law of independent assortment.

Ans: Each pair of alleles assorts independently of other pairs during gamete formation.

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Q9. [SLO 2] Define allele, homozygous, heterozygous.

Ans: Allele: alternative form of a gene at the same locus. Homozygous: two same alleles. Heterozygous: two different alleles.

Q10. [SLO 2] Differentiate genotype and phenotype.

Ans: Genotype: the pair of alleles. Phenotype: the visible expression of the trait.

Q11. [SLO 2] Differentiate monohybrid and dihybrid cross.

Ans: Monohybrid: parents differ in one trait (3:1 in F2). Dihybrid: two traits (9:3:3:1 in F2).

Q12. [SLO 2] Differentiate first and second filial generation.

Ans: F1: offspring of the parents (P). F2: offspring from F1 x F1.

Q13. [SLO 2] What is the F2 genotypic ratio of a monohybrid cross?

Ans: 1 RR : 2 Rr : 1 rr (phenotypic ratio 3:1).

Q14. [SLO 2] What is the dihybrid ratio?

Ans: 9 : 3 : 3 : 1 (round yellow : round green : wrinkled yellow : wrinkled green).

Q15. [SLO 2] What is a Punnett square? Who devised it?

Ans: A grid summarizing the fusion of gametes in a cross; devised by R. C. Punnett.

Q16. [SLO 2] What is a test cross?

Ans: Crossing an individual of unknown genotype with a homozygous recessive; all dominant phenotype shows homozygous, 1:1 shows heterozygous.

Q17. [SLO 3] How is independent assortment related to meiosis?

Ans: Homologous pairs line up independently at metaphase I (random orientation), so maternal and paternal chromosomes are sorted independently into gametes.

Q18. [SLO 3] How many chromosome combinations are possible in human gametes?

Ans: 2^{23} = about 8 million.

Q19. [SLO 4] State limitations of independent assortment.

Ans: Applies only to diploid organisms; linked genes on the same chromosome do not assort independently; X-linked genes give a different pattern.

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Q20. [SLO 5] List three sources of genetic variation in sexual reproduction.

Ans: Crossing over (prophase I), random assortment of chromosomes (metaphase I) and random fusion of gametes.

Q21. [SLO 6] Define probability. How does it relate to genetics?

Ans: P = favourable cases / total cases. Segregation of alleles and their pairing at fertilization follow probability rules (e.g. 3/4 dominant in F2).

Q22. [SLO 6] What is the product rule?

Ans: The probability of independent events occurring together is the product of their individual probabilities (3/4 x 3/4 = 9/16).

Q23. [SLO 7] What are the exceptions to Mendel's laws?

Ans: Incomplete dominance, co-dominance, multiple alleles (also epistasis, polygenic inheritance, linkage and sex linkage).

Q24. [SLO 8] What is incomplete dominance? Give an example.

Ans: Heterozygote has a phenotype intermediate between parents: red x white four o'clock gives pink; F2 1:2:1.

Q25. [SLO 8] Why are capital letters with dashes used in incomplete dominance?

Ans: No allele is dominant or recessive, so they are written with capital letters distinguished by dashes or numbers.

Q26. [SLO 9] What is co-dominance? Give an example.

Ans: Both alleles express independently in the heterozygote: MN blood group (L^ML^N) and AB blood group.

Q27. [SLO 9] Differentiate incomplete dominance and co-dominance.

Ans: Incomplete: heterozygote shows an intermediate blended phenotype.

Co-dominance: heterozygote shows both parental phenotypes together.

Q28. [SLO 10] Define multiple alleles. How many does one individual carry?

Ans: More than two alleles of a gene at the same locus (from mutation); a diploid individual carries only two.

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Q29. [SLO 10] Why can multiple alleles give many phenotypes?

Ans: Different combinations of the many alleles in a pair give many genotypes and phenotypes (3 alleles give 6 genotypes and 4 ABO groups).

Q30. [SLO 10] Differentiate alleles and multiple alleles; genes and alleles.

Ans: Alleles: two forms of a gene. Multiple alleles: more than two forms. Gene: a unit of heredity; alleles are its alternative forms.

Q31. [SLO 11] State the alleles of ABO blood group.

Ans: I^A, I^B and i; I^A and I^B are co-dominant, both dominant over i.

Q32. [SLO 11] Who discovered ABO blood groups?

Ans: Karl Landsteiner in 1901.

Q33. [SLO 11] Describe the antigens of the ABO system.

Ans: A and B glycoprotein antigens on RBC: A = group A; B = group B; both = AB; none = O.

Q34. [SLO 11] Describe the antibodies of the ABO system.

Ans: Group A has anti-B; B has anti-A; AB none; O both anti-A and anti-B; formed without any stimulus from embryonic life.

Q35. [SLO 12] How does the AB group show co-dominance?

Ans: I^A and I^B express equally in the heterozygote, giving both A and B antigens.

Q36. [SLO 13] Name various human blood group systems.

Ans: ABO, MNS, Rhesus (Rh), Lutheran, Kell, Lewis, Duffy and Kidd (33 systems in all).

Q37. [SLO 14] Why is O negative the universal donor?

Ans: It has no A, B or Rh antigen to react with recipient antibodies.

Q38. [SLO 14] Why is AB positive the universal recipient?

Ans: It has no anti-A, anti-B or anti-Rh antibodies, so no donor blood reacts.

Q39. [SLO 14] Why is the recessive allele i frequent in population?

Ans: ii x ii and several heterozygous crosses give O offspring, so i is carried on and persists (O is 37-47% in many countries).

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Q40. [SLO 14] What is agglutination?

Ans: Clumping of RBCs when donor antigens react with recipient antibodies.

Q41. [SLO 15] What is the Bombay phenotype?

Ans: hh individuals lack H substance (precursor of A and B), so show O phenotype even with I^A or I^B.

Q42. [SLO 15] Name two minor blood group systems.

Ans: MNS and Kell (also Lutheran, Duffy, Kidd).

Q43. [SLO 16] What is Rh factor? Who named Rh system?

Ans: The D antigen on RBC; named after Rhesus monkey (Landsteiner, 1930s).

Q44. [SLO 16] What are the genotypes of Rh positive and Rh negative persons?

Ans: Rh+: DD or Dd. Rh-: dd.

Q45. [SLO 16] How does anti-Rh antibody production differ from ABO antibodies?

Ans: Anti-Rh needs a stimulus (exposure to Rh antigen); ABO antibodies form without stimulus.

Q46. [SLO 17] Why is Rh incompatibility a danger to foetus and mother?

Ans: Anti-Rh antibodies of a sensitized Rh- mother cross the placenta and haemolyse the Rh+ foetal RBCs: anaemia, jaundice, abortion or stillbirth.

Q47. [SLO 18] What is erythroblastosis foetalis?

Ans: Haemolytic disease of the newborn: anaemic foetus releases immature erythroblasts after destruction of its RBCs by maternal anti-Rh antibodies.

Q48. [SLO 18] What is an erythroblast?

Ans: A type of red blood cell that still keeps its nucleus.

Q49. [SLO 19] How is erythroblastosis prevented?

Ans: By giving the Rh- mother RhoGAM (anti-Rh immunoglobulin) at 28 weeks and within 72 hours of delivery of an Rh+ baby.

Q50. [SLO 20] Define polygenic inheritance and epistasis.

Ans: Polygenic: a trait controlled by several gene pairs with additive effects (continuous variation). Epistasis: a gene at one locus masks the effect of a gene at

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another locus.

Q51. [SLO 20] How do polygenes and multiple alleles differ?

Ans: Polygenes: alleles of many genes at different loci add up. Multiple alleles: many alleles of one gene at one locus.

Q52. [SLO 21] Why is wheat grain colour polygenic?

Ans: Three gene pairs (A, B, C) each add red pigment; F2 shows seven shades 1:6:15:20:15:6:1 depending on the number of pigment alleles.

Q53. [SLO 21] Which pigment controls human skin colour? How many genes?

Ans: Melanin; at least three genes A, B, C with seven shades from aabbcc to AABBCC.

Q54. [SLO 22] List five polygenic traits of humans.

Ans: Height, skin colour, intelligence, blood pressure, risk of heart disease or diabetes (also autism, longevity).

Q55. [SLO 23] Differentiate dominance and epistasis.

Ans: Dominance: between alleles of one gene (same locus). Epistasis: between genes at different loci, one masking the other.

Q56. [SLO 23] What are epistatic and hypostatic genes?

Ans: Epistatic gene suppresses the effect of a gene at another locus; the suppressed gene is hypostatic.

Q57. [SLO 23] What ratio is seen in Labrador coat colour?

Ans: 9 black : 3 chocolate : 4 yellow (recessive epistasis by ee).

Q58. [SLO 23] What is the foxglove ratio?

Ans: 12 : 3 : 1 (dominant epistasis; W restricts colour).

Q59. [SLO 24] What is gene linkage and linkage group?

Ans: Linked genes on the same chromosome inherited together. All linked genes of one homologous pair form a linkage group (number = haploid chromosome number).

Q60. [SLO 24] How is linkage detected?

Ans: By a dihybrid test cross: 1:1:1:1 = no linkage; more parental types = incomplete linkage; only parental = complete linkage.

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Q61. [SLO 24] Explain crossing over.

Ans: Exchange of parts between maternal and paternal chromatids of paired homologues in prophase I, forming recombinant chromatids and variation.

Q62. [SLO 25] How does gene linkage counter independent assortment?

Ans: Linked genes stay together, so no recombinants form and 9:3:3:1 is modified (to 3:1 in Morgan's Drosophila cross); crossing over then modifies the progeny.

Q63. [SLO 26] Why is a large number of progeny needed to observe linkage?

Ans: A larger sample gives more statistical power to detect deviation from independent assortment; small samples show chance mixing.

Q64. [SLO 26] What is a map unit?

Ans: Distance between linked genes: one map unit = 1% recombination (crossing over).

Q65. [SLO 27] Describe the XX-XY pattern of sex determination.

Ans: Female XX (homogametic, X eggs only); male XY (heterogametic, X and Y sperms equally); sperm decides sex; ratio 1:1.

Q66. [SLO 27] Why is the human male heterogametic?

Ans: He produces two kinds of sperms, X-bearing and Y-bearing, in equal numbers.

Q67. [SLO 27] What is the SRY gene?

Ans: Sex-determining region Y gene on the Y chromosome that makes the individual male.

Q68. [SLO 28] How can sex be identified from a karyotype?

Ans: Females have two similar X chromosomes (XX); males have X and a smaller Y (XY) in the 23rd pair.

Q69. [SLO 29] Differentiate sex determination in Drosophila and human.

Ans: Both are XX-XY; Drosophila has 8 chromosomes (4 pairs) with hooked Y, humans 46 (23 pairs) and Y carries SRY.

Q70. [SLO 30] What is sex linkage? Differentiate X-linked and Y-linked inheritance.

Ans: Traits controlled by sex chromosome genes. X-linked: gene on X (both sexes, males show recessive easily). Y-linked (holandric): gene only on Y, only males, father

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to son.

Q71. [SLO 30] What are homogametic, hemizygous and holandric?

Ans: Homogametic: one type of gamete (XX). Hemizygous: single allele (males for X-linked genes). Holandric: Y-linked genes.

Q72. [SLO 31] What did Morgan conclude from the white-eye cross?

Ans: The white-eye gene is recessive and on the X chromosome; Y carries no allele.

Q73. [SLO 33] Why do X-linked recessive disorders affect males more?

Ans: Males have one X, so one recessive allele shows; females need two copies.

Q74. [SLO 34] Name sex-linked disorders of man and Drosophila.

Ans: Man: red-green colour blindness, haemophilia A and B. Drosophila: white eye.

Q75. [SLO 34] Under what circumstances can a father and son both be haemophilic?

Ans: When the mother is a carrier (X^HX^h) or affected and the son inherits her X^h; the father passes only Y to his son.

Q76. [SLO 34] What is dichromacy and monochromacy?

Ans: Dichromacy: one primary colour unseen (protanopia, deuteranopia, tritanopia). Monochromacy: only one colour seen (true colour blindness).

Q77. [SLO 34] Why is red-green colour blindness more common in men?

Ans: The genes for red and green opsin are on the X chromosome; a man with one recessive allele is affected, a woman needs two.

Q78. [SLO 34] What are sex limited and sex influenced traits?

Ans: Sex limited: expressed in one sex only. Sex influenced: dominant in one sex and recessive in the other, as baldness.

Q79. [SLO 35] What is amniocentesis?

Ans: Withdrawal of about 20 ml amniotic fluid with a needle (ultrasound guided) to culture foetal cells and examine chromosomes, e.g. Down syndrome.

Q80. [SLO 35] What is chorionic villus sampling?

Ans: Taking a sample of placental chorionic villi through the cervix or abdomen for early karyotyping and genetic testing.

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Chapter 12: Long Questions with Answers (SLO based)

Q1. [SLO 2] Explain Mendel's law of inheritance in a monohybrid cross and interpret the results.

Ans:

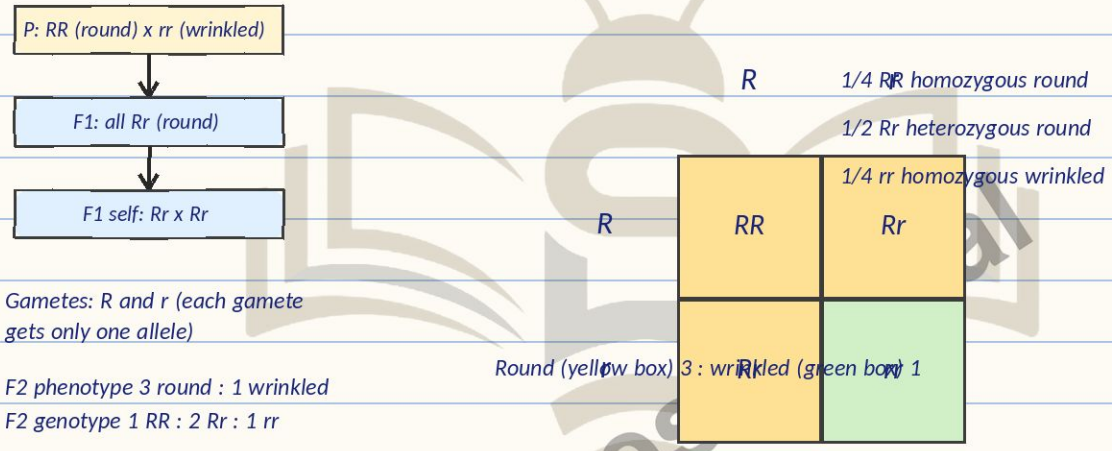


Fig: Monohybrid cross (Law of segregation)

- *Pure round (RR) x wrinkled (rr): F1 all round (Rr). Selfing gives F2 3 round : 1 wrinkled, genotype 1:2:1. Interpretation: each trait controlled by a pair of factors (alleles), one from each parent; dominant expressed in F1; alleles segregate in gamete formation (law of segregation); fertilization random.*

Q2. [SLO 2] Explain Mendel's dihybrid cross and law of independent assortment.

Ans:

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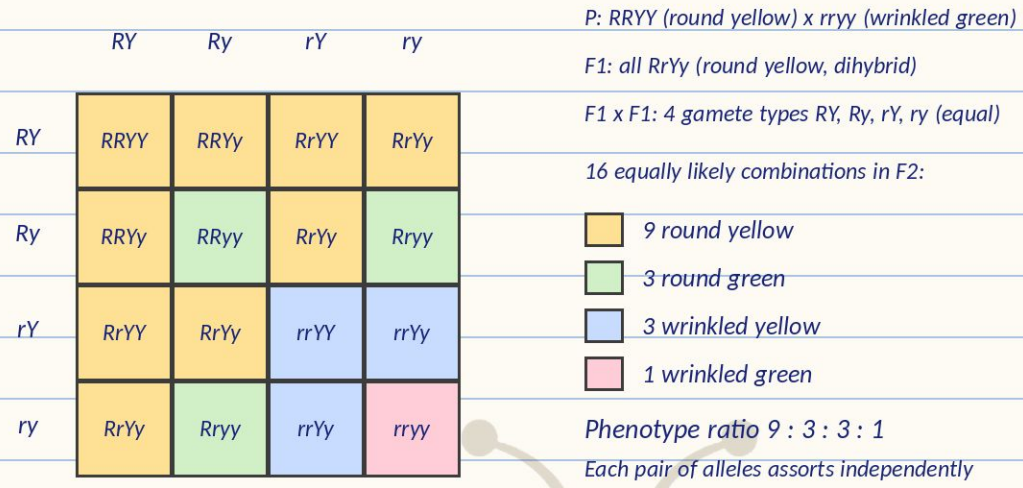


Fig: Dihybrid cross (Law of independent assortment)

- *RRYY x rryy gives F1 RrYy (round yellow); F2 9:3:3:1 from 16 combinations of four gamete types. Each pair of alleles assorts independently of the other (genes on different chromosomes).*

Q3. [SLO 3, 4, 5] Describe law of independent assortment under limitations and usefulness; show that independent assortment leads to variation in gametes.

Ans:

- *Relation to meiosis: homologous pairs orient randomly at metaphase I; 2^{23} = 8 million combinations in human gametes.*
- *Limitations: diploid only; linked genes; X-linked genes.*
- *Usefulness: source of variation with crossing over and mutation; new combinations for adaptation; random fertilization gives different zygotes.*

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Q4. [SLO 6] What is probability? How can you evaluate that inheritance of genes and their mixing during fertilization is based on mathematical probabilities?

Ans:

- *P = a/n. Mendel's monohybrid F2: dominant 3/4, recessive 1/4; dominant in F1 100%.*
- *Product rule: P(round yellow) = 3/4 x 3/4 = 9/16; P(wrinkled green) = 1/4 x 1/4 = 1/16. Gamete formation and random fertilization behave like coin toss, so observed ratios approach expected ratios in large numbers.*

Q5. [SLO 8, 9] Explain incomplete dominance and co-dominance with examples; differentiate them.

Ans:

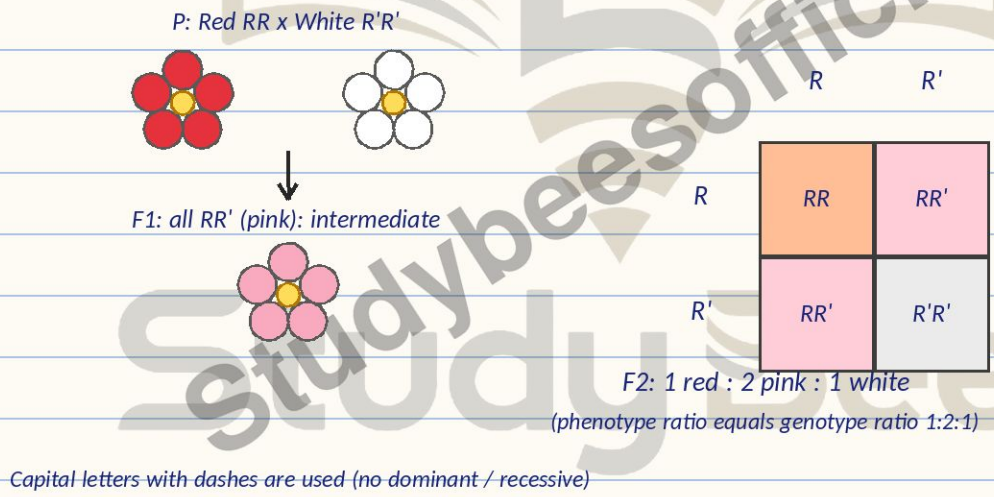


Fig: Incomplete dominance in four o'clock (Mirabilis jalapa)

Incomplete dominance	***Co-dominance***
Blend, intermediate phenotype	*Both phenotypes expressed*
Four o'clock, pink flower	*MN and AB blood groups*

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Q6. [SLO 10-12] Describe multiple alleles with reference to ABO blood group genetics.

Ans:

- Gene I with three alleles I^A, I^B, i; I^A = A antigen, I^B = B antigen, i = none; I^A and I^B co-dominant; both dominant over i.

Group	Genotype / antigen / antibody
A	I^AI^A, I^Ai; A; anti-B
B	I^BI^B, I^Bi; B; anti-A
AB	I^AI^B; A and B; none
O	ii; none; anti-A and anti-B

Q7. [SLO 13-15] Name various blood group systems and investigate why O negative is the universal donor and AB positive the universal recipient.

Ans:

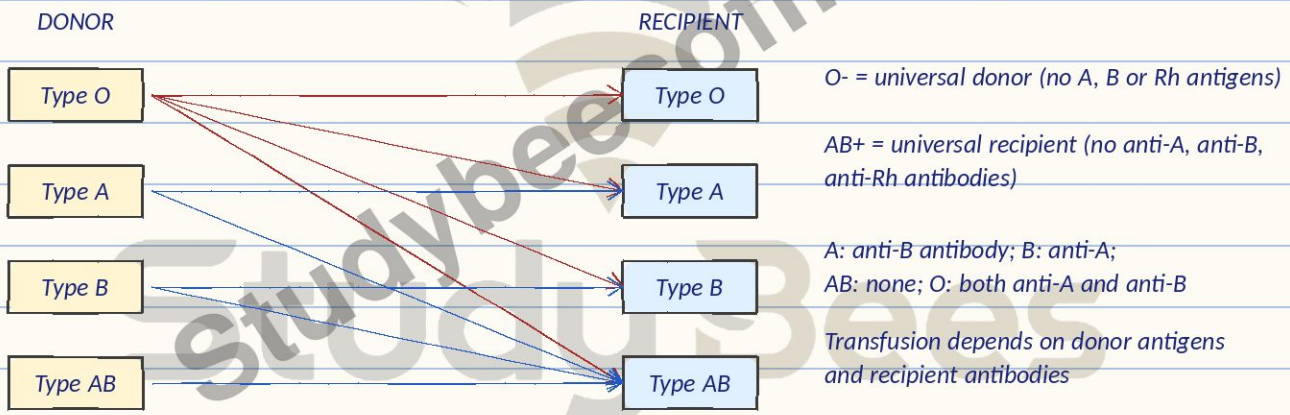


Fig: ABO transfusion compatibility

- Systems: ABO, MNS, Rh, Lutheran, Kell, Lewis, Duffy, Kidd (33 in all). O negative lacks A, B and D antigens; AB positive lacks anti-A, anti-B, anti-Rh.

Q8. [SLO 16-19] Describe the Rh blood group system and explain erythroblastosis foetalis in light of antigen-antibody reaction; suggest measures to counter it.

Ans:

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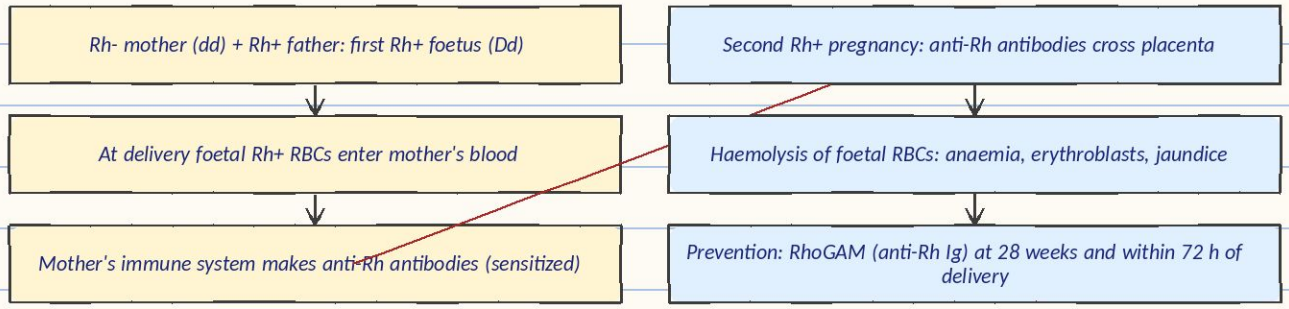


Fig: Rh incompatibility and erythroblastosis foetalis

- *Rh+ (DD, Dd) have D antigen; Rh- (dd) do not. Anti-Rh made only after exposure. Rh- mother carrying Rh+ foetus is sensitized at delivery; later Rh+ foetus attacked by anti-Rh crossing placenta: haemolysis, anaemia, erythroblasts, jaundice, stillbirth.*
- *Prevention: RhoGAM at 28 weeks, within 72 hours of delivery and after miscarriage, amniocentesis or bleeding; replace the baby's blood with Rh- blood.*

Q9. [SLO 20-22] Describe polygenic inheritance using wheat grain colour and human skin colour; list five human polygenic traits.

Ans:

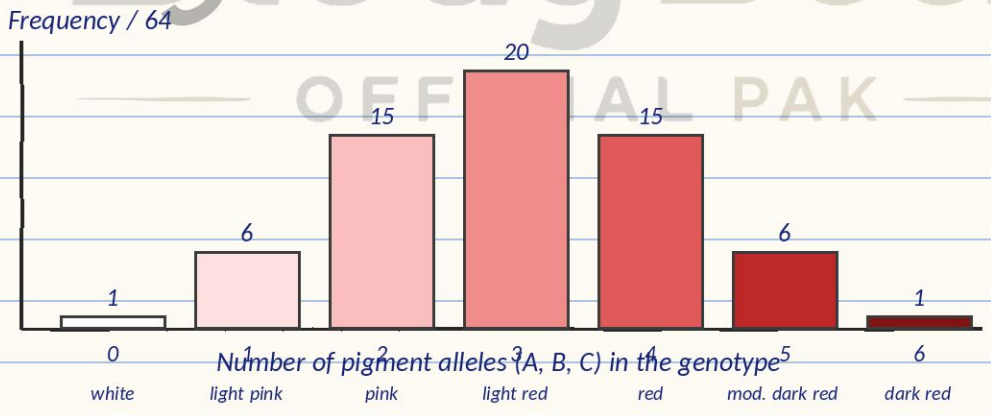


Fig: Polygenic inheritance of wheat grain colour (also human skin colour)

- *Wheat: three gene pairs; F2 1:6:15:20:15:6:1 (seven shades). Skin colour:*

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melanin controlled by A, B, C; seven shades; most AaBbCc.

- *Human polygenic traits: height, skin colour, intelligence, blood pressure, heart disease risk (also diabetes, autism, longevity).*

Q10. [SLO 23] Define epistasis; describe it with Labrador coat colour and foxglove; justify modified Mendelian ratios.

Ans:

	BE	Be	bE	be
BE	BBEE	BBEe	BbEE	BbEe
Be	BBEe	BBee	BbEe	Bbee
bE	BbEE	BbEe	bbEE	bbEe
be	BbEe	Bbee	bbEe	bbee

BBEE (black) x bbee (yellow): F1 all BbEe (black)

BbEe x BbEe gives three coat colours:

- *9 black (B_E_)*
- *3 chocolate (bbE_)*
- *4 yellow (__ee)*

Ratio 9 : 3 : 4 (recessive epistasis)

ee is epistatic: masks B and b (different loci)

Foxglove (dominant epistasis): 12 : 3 : 1

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Fig: Epistasis - coat colour in Labrador retriever

- *Labrador: ee epistatic to B / b gives 9:3:4. Foxglove: dominant W epistatic gives 12:3:1. Modified because genes at different loci interact: the normal 9:3:3:1 combines classes.*

Q11. [SLO 24-26] Describe gene linkage and crossing over; explain how linkage counters independent assortment and why large progeny are needed.

Ans:

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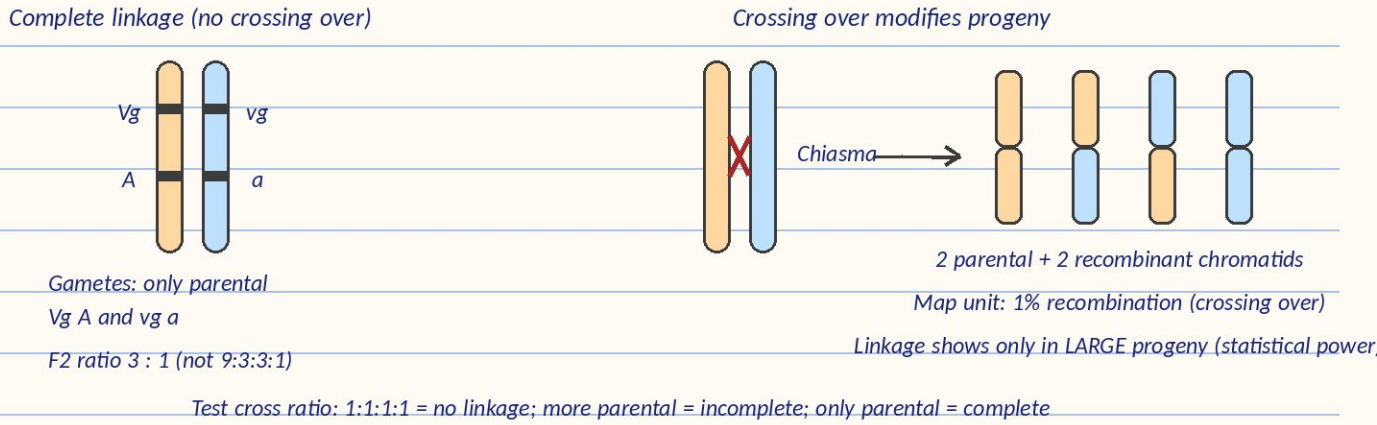


Fig: Gene linkage and crossing over (Morgan, Drosophila)

- *Linked genes (same chromosome) inherited together; test cross detects linkage; Morgan's Drosophila F2 gave 3:1; crossing over in prophase I makes recombinants and modifies progeny; map unit = 1% crossing over; large progeny give statistical power.*

Q12. [SLO 27-29] Explain the mechanism of sex determination in mammals and solve sex determination problems.

Ans:

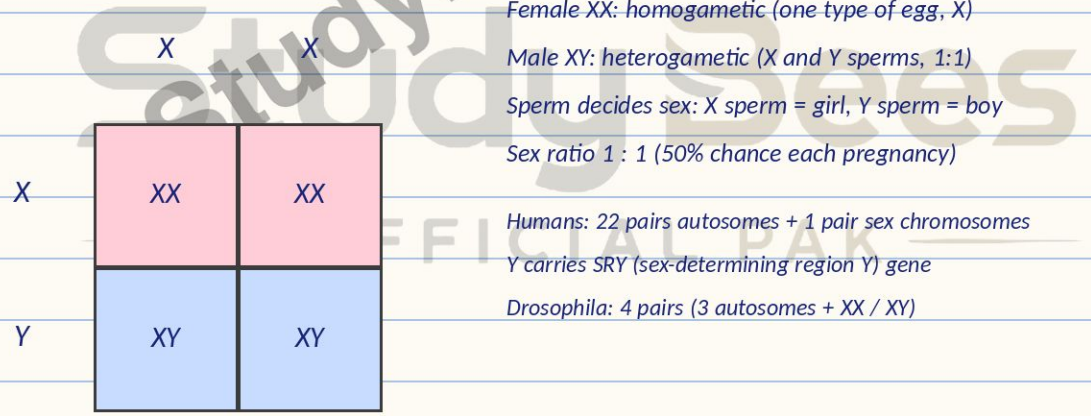


Fig: XX-XY sex determination

- *Karyotype: 22 autosomal pairs + XX (female) or XY (male). Sperm determines sex; ratio 1:1. SRY on Y.*

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Q13. [SLO 30-34] Describe sex linkage: eye colour in Drosophila, X-linked and Y-linked traits, haemophilia and colour blindness.

Ans:

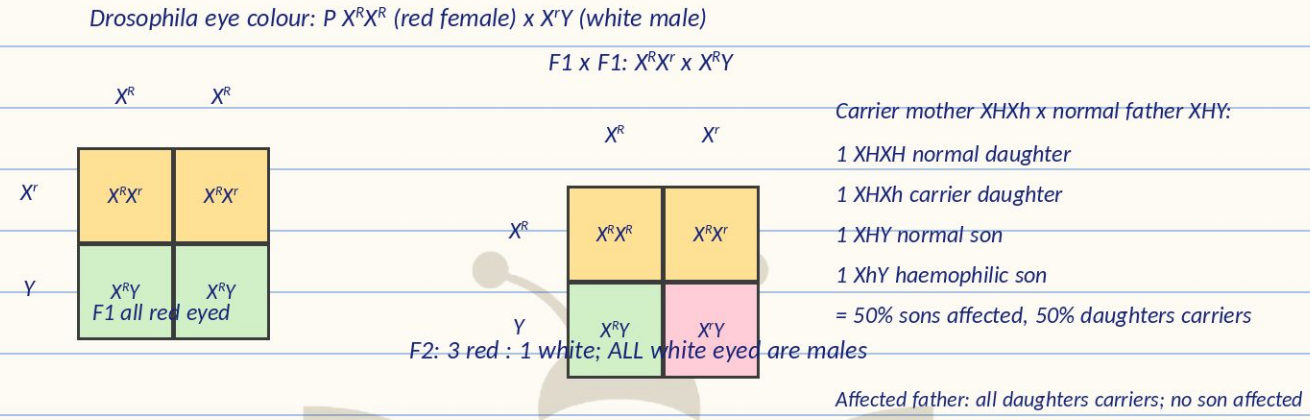


Fig: Sex-linked (X-linked recessive) inheritance

- *X-linked recessive: males affected; carrier mothers; zigzag; Y-linked: father to son only (hypertrichosis, webbed toes). Haemophilia A/B and colour blindness are X-linked recessive; haemophilia C autosomal.*

Q14. [SLO 35] Explain amniocentesis and chorionic villus sampling.

Ans:

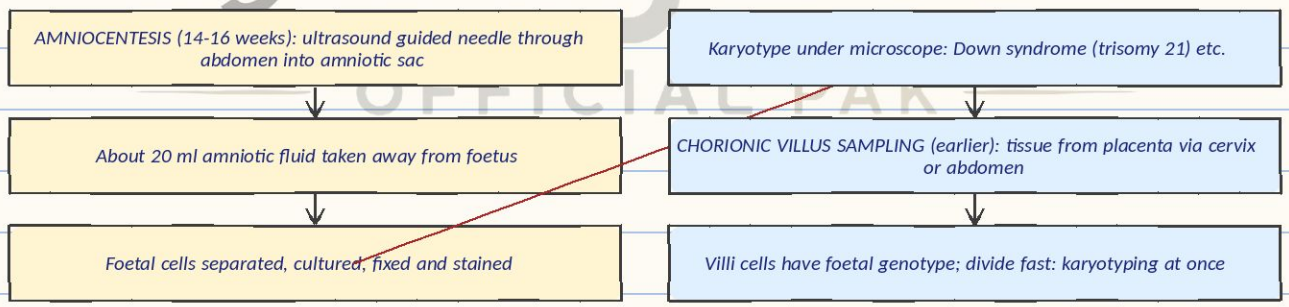


Fig: Embryonic screening techniques

- *Amniocentesis: needle into amniotic sac, 20 ml fluid, cells cultured, karyotype*

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(Down syndrome). CVS: placental villi taken via cervix or abdomen; early karyotype.

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